UDACITY
Part of Accenture

Calculation of Basketball Scoring Probabilities

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Agenda

Results of basketball scoring calculations

- Overall Shooting Statistics
- Probabilities
- Conditional Probabilities - Future
- Conditional Probabilities - Retrospective

Overall Shooting Statistics

Metric	Value
Total shots attempted	1,434
Total makes (result = True)	703
Total misses (result = False)	731
P(Make) = 703 / 1434	0.4902 (49.02%)
P(Miss) = 731 / 1434	0.5098 (50.98%)
Three-point attempts	786
Two-point attempts	648
P(3-pointer) = 786 / 1434	0.5481 (54.81%)
P(2-pointer) = 648 / 1434	0.4519 (45.19%)

Probabilities: Binomial Distribution

Using the overall make probability p = 0.4902 and the three-point rate p_3 = 0.5481, we apply the Binomial Distribution formula:

$$\mathbf{P(X = k) = C(n, k) \cdot p^k \cdot (1-p)^{n-k}}$$

Question A: Probability of making exactly 3 of the next 4 shots

X ~ Binomial(n = 4, p = 0.4902)

$P(X = 3) = C(4, 3) \times 0.4902^3 \times 0.5098^1$

$= 4 \times 0.117757 \times 0.5098$

$= 4 \times 0.060033$

Result: P(X = 3) = **0.2402 (24.02%)**

Question B: Probability that exactly 4 of the next 5 shots are three-pointers

X ~ Binomial(n = 5, p = 0.5481)

$P(X = 4) = C(5, 4) \times 0.5481^4 \times 0.4519^1$

$= 5 \times 0.090128 \times 0.4519$

$= 5 \times 0.040739$

Result: P(X = 4) = **0.2039 (20.39%)**

Probabilities: Binomial Distribution

Assumption	Description	Satisfied?
Fixed n trials	Number of shots is set before the event.	Yes
Binary outcomes	Each shot is either a Make or a Miss.	Yes
Constant p	P(Make) is the same for every shot.	Partially – shot difficulty varies
Independence	Each shot outcome does not affect the next.	Questionable – hot-hand / fatigue effects

- The Binomial model is therefore a useful first-order approximation rather than an exact model.

Conditional Probabilities - Future

Bayes Theorem

Condition	P(Make | Shot Type)	P(Lead | Shot Type)
Given a 3-pointer	0.4186 (41.86%)	0.4936 (49.36%)
Given a 2-pointer	0.5772 (57.72%)	0.5154 (51.54%)

Conditional Probabilities - Future

Bayes' Theorem: $P(A \mid B) = P(B \mid A) \times P(A) / P(B)$

THREE-POINTER

i) P(Make | 3-pointer) — direct conditional probability

Known counts: Makes AND 3pt = 329; Total 3pt attempts = 786

$P(\text{Make} \cap \text{3pt}) = 329 / 1434 = 0.2294$

$P(\text{Make} \mid \text{3pt}) = P(\text{Make} \cap \text{3pt}) / P(\text{3pt})$

$= 0.2294 / 0.5481 = 0.4186$

Result: P(Make | 3-pointer) = 0.4186 (41.86%)

Conditional Probabilities - Future

ii) P(Lead = True | 3-pointer) — Bayes' Theorem

Let A = Lead, B = 3-pointer

P(Lead) = 722 / 1434 = 0.5035

P(3pt | Lead) = 388 / 722 = 0.5374

P(Lead | 3pt) = P(3pt | Lead) × P(Lead) / P(3pt)

= 0.5374 × 0.5035 / 0.5481

= 0.2705 / 0.5481

Result: P(Lead | 3-pointer) = 0.4936 (49.36%)

Conditional Probabilities - Future

TWO-POINTER

i) P(Make | 2-pointer)

Known counts: Makes AND 2pt = 374; Total 2pt attempts = 648

$P(\text{Make} \cap \text{2pt}) = 374 / 1434 = 0.2608$

$P(\text{Make} \mid \text{2pt}) = P(\text{Make} \cap \text{2pt}) / P(\text{2pt})$

$= 0.2608 / 0.4519 = 0.5772$

Result: P(Make | 2-pointer) = 0.5772 (57.72%)

ii) P(Lead = True | 2-pointer) — Bayes' Theorem

$P(\text{2pt} \mid \text{Lead}) = 334 / 722 = 0.4626$

$P(\text{Lead} \mid \text{2pt}) = P(\text{2pt} \mid \text{Lead}) \times P(\text{Lead}) / P(\text{2pt})$

$= 0.4626 \times 0.5035 / 0.4519$

$= 0.2329 / 0.4519$

Result: P(Lead | 2-pointer) = 0.5154 (51.54%)

Conditional Probabilities - Future

Condition	P(Make | Shot Type)	P(Lead | Shot Type)
Given a 3-pointer	0.4186 (41.86%)	0.4936 (49.36%)
Given a 2-pointer	0.5772 (57.72%)	0.5154 (51.54%)

Conditional Probabilities - Retrospective

Given that Steph just made a shot, what is the probability it was a three-pointer or a two-pointer? We use Bayes' Theorem in its standard form:

$$P(A \mid B) = P(B \mid A) \times P(A) / P(B)$$

Notation mapping

- "Made three-pointer" → P(Made, 3pt)
- "Three-pointer given a make" → P(3pt | Made)
- "Make" → P(Made) = 0.4902

Conditional Probabilities - Retrospective

P(3-pointer | Made)

Bayes: P(3pt | Made) = P(Made | 3pt) × P(3pt) / P(Made)

Step 1 – Identify terms:

P(Made | 3pt) = 329 / 786 = 0.4186

P(3pt) = 786 / 1434 = 0.5481

P(Made) = 703 / 1434 = 0.4902

Step 2 – Numerator: 0.4186 × 0.5481 = 0.2294

Step 3 – Divide: 0.2294 / 0.4902 = 0.4680

Result: P(3-pointer | Made) = 0.4680 (46.80%)

Conditional Probabilities - Retrospective

P(2-pointer | Made)

Bayes: P(2pt | Made) = P(Made | 2pt) × P(2pt) / P(Made)

Step 1 – Identify terms:

P(Made | 2pt) = 374 / 648 = 0.5772

P(2pt) = 648 / 1434 = 0.4519

P(Made) = 703 / 1434 = 0.4902

Step 2 – Numerator: 0.5772 × 0.4519 = 0.2608

Step 3 – Divide: 0.2608 / 0.4902 = 0.5320

Result: P(2-pointer | Made) = 0.5320 (53.20%)

Verification by direct counting:

P(3pt | Made) = 329 / 703 = 0.4680

P(2pt | Made) = 374 / 703 = 0.5320

Sum = 0.4680 + 0.5320 = 1.0000 (probabilities are exhaustive and mutually exclusive)

Conditional Probabilities - Retrospective

Retrospective Question	Bayes Result
P(3-pointer | just Made a shot)	0.4680 (46.80%)
P(2-pointer | just Made a shot)	0.5320 (53.20%)